

Physical activity energy expenditure has not declined since the 1980s and matches energy expenditures of wild mammals.

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OBJECTIVE: Obesity results from protracted energy imbalance. Whether this comprises excessive energy intake, lowered physical activity or both, remains disputed. DESIGN: Physical activity energy expenditure, evaluated in three different ways from daily energy expenditure (DEE) measured using doubly labelled water, was examined for trends over time. Data included subjects in Europe (Maastricht, the Netherlands) and North America extending back to the 1980s. These data were compared with measures from the third world, and measures made on wild terrestrial mammals. RESULTS: Physical activity expenditure in Europe (residual of the regression of DEE on basal energy expenditure (BEE)) has slightly but significantly increased since the 1980s. There was no trend over time in physical activity level ($PAL = DEE/BEE$), or in the residual variance in DEE once mass, sex and age were accounted for. This latter index of physical activity expenditure also significantly increased over time in North America. DEE of individuals in Europe and North America was not significantly different from individuals measured in the third world. In wild terrestrial mammals, DEE mostly depended on body mass and ambient temperature. Predicted DEE for a 78 kg mammal living at 20 degrees C was 9.2 MJ per day (95% CI: 7.9-12.9 MJ per day), not significantly different from the measured DEE of modern humans (around 10.2-12.6 MJ per day). CONCLUSION: As physical activity expenditure has not declined over the same period that obesity rates have increased dramatically, and daily energy expenditure of modern man is in line with energy expenditure in wild mammals, it is unlikely that decreased expenditure has fuelled the obesity epidemic.